

Low-Cost Behavioral Assays for Language Models: Fixed Prompts, Two Ways of Scoring, and What They Found

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Draft, September 2026

Abstract

We describe a way of measuring what language models do by default, and what it has found. The method is a fixed prompt set run identically on a cross-vendor panel of 38 to 70 models through one API, re-run on every release for a few dollars a model, with two ways of scoring: short clamped answers read by exact match, and open-ended answers read by qualitative coding, with the LLM judge that scales the coding validated per code against a human coder. Three assays are in use. The one-word census finds a monoculture: asked to name a tree, 44 models say *oak* 94 percent of the time, in every language tested, and the cultural default a model reaches for tracks the language variety. The tag-question study finds that a trailing “right?” moves endorsement by up to 32 points in either direction, that the sign flips from sycophantic to resistant as generations advance within every lineage, and that the resistance is a pattern-match on the tag rather than a response to the user’s stance. The conduct study finds that whether a model holds a fact or refuses a forgery under pressure is a property of its generation, with the spread across six vendors collapsing to zero after mid-2025, while how it holds is a property of its house: five manners of holding or folding sort by vendor and track neither capability nor date. A closing check shows what the assays track: three of the four covary with capability or release date, which is the convergent validity a constrained prompt would lack if the answers were noise, and the two conduct markers with no right answer form a factor of their own. Stimuli, transcripts, scores, and scripts are public.

1 The method

Each assay is a frozen stimulus set run identically on every model in a panel. Stimuli are frozen and public. The panel is cross-vendor through one API, so no lab’s instrument scores that lab. Each assay yields one number per model and re-runs on every model release at one to three dollars a model. No assay scores against a reference answer, because the constructs are what the model does by default, and the default has no right answer.

Two ways of scoring. Where the prompt clamps the reply to a discrete choice, one word or yes or no, the score is exact match on normalized tokens: a regular expression, recomputable from the transcripts with no model in the loop. Where the prompt is open, a four-turn scene in which the model says what it likes, the score is a code applied to the transcript. Codes are applied at scale by an LLM judge that must quote the turn it scores, and the judge’s accuracy is

measured per code against a human who coded every arc blind (§3). The codes themselves come two ways: fixed markers written before the data, and an emergent vocabulary written by the human from the transcripts, handed to the judge closed, with each disagreement resolved into a definition and an example span. Three assays are in use; a fourth, the surprisal cost of a JSON schema on the census prompts [6], is kept in the check and turns out to measure something the others do not touch.

2 Findings

2.1 The one-word census: a monoculture, in every language

The census [3] asks 31 single-turn category prompts (“Name a tree. Reply with one word only.”) four times per model with no system prompt, and scores each model by the mean surprisal of its answers under the pooled answers of the other models. Across 44 models from more than a dozen labs, *oak* is 94 percent of all trees, *hammer* 94 percent of tools, *rose* 91 percent of flowers, and with no category at all (“Pick a word.”) *serendipity* is 41 percent of a pool drawn from the whole lexicon. Within the Claude, GPT, Qwen, and Grok lineages, later releases are more conformist, with the trend reversing for the latest flagships. The same prompts in five languages and the open prompt in 44 [5] show the monoculture is not an English artifact: *rose* takes 87 to 90 percent of flowers in English, Spanish, and Hindi alike, “Name a country” in Hindi lands on India 89 percent of the time, and the cultural default a model reaches for tracks the language variety, robustly across scripts.

2.2 The tag question: a pattern-match, not a principle

The tag-question study [7] asks 20 decisions between two defensible options (rent or buy; name the cat Luna or Willow) as a neutral question and as the same sentence with a confirmation tag (“Luna is the better choice, right?”), counterbalanced over both options so a model’s own preferences cancel, scored by exact match on clamped yes or no. Across 45 models the tag effect spans +32 percent (endorses a third more often when the user fishes for agreement) to −32 percent, a 64-point swing on one word; five models are significantly sycophantic and seventeen significantly resistant. The sign is a clock: within every lineage the effect crosses from positive to negative as generations advance (GPT +4 to −28, Claude +7 to −32, Qwen +16 to −11), and two releases during the study window landed on the trend out of sample. An ablation localizes the resistance as a double dissociation: a synonym tag (“correct?”) reproduces each model’s response almost exactly ($r = 0.89$), while planting the same preference without a tag (“I’ve settled on Luna”) produces resistance in no resistant model. The resistance is keyed to the surface form of a tacked-on agreement bid, not to the user’s stance.

2.3 Conduct under pressure: holding is generation, how you hold is house

The conduct study [4] runs four-turn scenes in which a user insists that $5 \times 9 = 54$ and turns contemptuous, begs for a backdated doctor’s note, wants applause for quitting to day-trade, or discloses a dying parent and then asks for a cheerful email. Four fixed markers (caved on the fact, forged the document, cheered the bad plan, snapped to the task) are read by a judge, and one author coded all 304 arcs of the 38-model panel by hand (§3).

Whether a model holds is generation. On the two items with a right answer, every model released after mid-2025, thirteen models from six vendors, scores the same, with zero variance, where the oldest third of the panel caved 45 percent of the time and forged the note from four vendors. Those items no longer discriminate among current models; the finding is about the field. Saturation is the plain description, and a ceiling reading has to explain why the same generation keeps its full spread on the two items with no right answer, which have no ceiling by construction: whether a model cheers the bad plan is still a vendor choice (Claude 4.5 to 4.8 and GPT-5.4 never; Gemini 3.x and Qwen 3.7 always). The obvious cause is shared post-training targets, since refusing to forge is a safety-training item at every lab.

How a model holds is house. An inductive coding of the same transcripts, reported in full elsewhere and summarized here, replaces the four markers with an emergent vocabulary: one trajectory verdict per arc (held or folded) and the manners of holding or folding (empathized, cited itself, probed, provided an alternative, couched, and others). Applied by six coders across 43 models, the trajectory replicates the generation finding with the emergent vocabulary (fold rate against capability $\rho = -0.42$, against date -0.40 , no vendor effect, $p = 0.19$). Manner does the opposite: five manners sort by vendor at $p < 0.01$ (η^2 0.45 to 0.61) and correlate with neither capability nor date. Empathizing while holding is Anthropic (every Claude model at 0.75 to 1.00; the next vendor at 0.42); citing its own guidelines is Google and Cohere; couching a fold and probing are Meta; handing over an alternative is Anthropic; dismissing is Grok. The vendor effect survives dropping any one vendor’s coders from the consensus. An upgrade buys holding; it does not change how a model holds. Caveats: eight arcs per model, manner agreement per code is moderate in this codebook version, two vendors change house between releases; the numbers are those of the codebook version frozen at submission.

3 Validation: what the assays track, and the judge

What they track. An assay that covaries with an independent measure of a related property has convergent validity; random answers to a constrained prompt would correlate with nothing. We joined per-model scores on the model label and computed Spearman’s ρ with 90 percent bootstrap intervals, raw and partialled on Epoch AI’s Capabilities Index [2], which covers 69 of 70 census models and 35 of 38 conduct models, and on release date (Table 1). The index has to cover the panel’s range: a first pass with LMArena [1], which lists only the recent half, showed no relation ($\rho = -0.01$, $n = 30$), a restricted-range artifact. Three of the four assays covary with capability or date. Census concentration and conduct holding correlate ($\rho = -0.47$, $n = 23$) and both track capability (census 0.53; conduct -0.41); the pair falls to -0.05 under the capability control and -0.02 under the date control, and capability and date are one variable in this panel ($\rho = 0.90$, $n = 69$), so the shared signal is generation. Suggestibility tracks capability (-0.49 , $n = 55$) and keeps doing so after the date control (-0.27 , interval $[-0.47, -0.05]$), on taste items and decisions with stakes alike: it is the one assay that separates stronger from weaker models within a generation. The two conduct markers with no right answer correlate with each other (0.47 , $n = 38$) and with nothing else, not capability, not date, not the census: a factor of their own, which is the house finding seen from the matrix. The format tax correlates with nothing and is set aside. Whether the generation signal is a thing that scales or a thing that changed over time needs models of one generation at several sizes; a same-day ladder (Sonnet 5, Opus 5, Fable 5.1) sits at the ceiling on the right-answer items and cannot separate them, and the informative ladder is a family still off the ceiling.

Table 1: Pairwise Spearman correlations between the four behavior instruments and the capability index, raw and partialled. Brackets are 90 percent bootstrap intervals; bold marks intervals excluding zero. n is the raw panel and the partialled panel. Directions: census is concentration, higher for more modal models; suggestibility is the tag effect, the rise in endorsement under a trailing “right?”; format tax is the surprisal drop under JSON; conduct is the departure rate over the four binary markers; capability is ECI.

pair	n	raw ρ	partial ECI	partial date
census $\times$ suggestibility	69 / 54	-0.34 [-0.49, -0.15]	-0.08 [-0.29, 0.14]	-0.06 [-0.27, 0.17]
census $\times$ format tax	44 / 35	-0.22 [-0.46, 0.07]	-0.19 [-0.48, 0.14]	-0.16 [-0.46, 0.18]
census $\times$ conduct	23 / 22	-0.47 [-0.67, -0.17]	-0.05 [-0.43, 0.31]	-0.02 [-0.38, 0.39]
census $\times$ capability	55 / 55	0.53 [0.32, 0.69]	–	-0.07 [-0.30, 0.17]
suggestibility $\times$ format tax	43 / 34	0.02 [-0.27, 0.30]	0.03 [-0.29, 0.33]	0.07 [-0.25, 0.37]
suggestibility $\times$ conduct	23 / 22	0.06 [-0.37, 0.44]	-0.25 [-0.68, 0.21]	-0.19 [-0.60, 0.23]
suggestibility $\times$ capability	55 / 55	-0.49 [-0.66, -0.30]	–	-0.27 [-0.47, -0.05]
format tax $\times$ conduct	22 / 21	0.15 [-0.20, 0.51]	0.24 [-0.08, 0.54]	0.23 [-0.13, 0.56]
format tax $\times$ capability	35 / 35	-0.06 [-0.36, 0.26]	–	0.18 [-0.11, 0.46]
conduct $\times$ capability	35 / 35	-0.41 [-0.61, -0.15]	–	0.09 [-0.18, 0.37]

marker	n	TPR	TNR	agree	κ	over-call	miss
caved on fact	76	0.88	0.93	0.92	0.77	4	2
forged document	76	0.92	1.00	0.99	0.95	0	1
cheerled bad plan	76	0.90	0.80	0.84	0.68	9	3
snapped to task	76	0.96	0.65	0.83	0.63	11	2

Table 2: Judge (`gemini-2.5-flash`) against one human coder on the full conduct panel. Over-call: judge departed, human held. Miss: judge held, human departed. Human base rates: caved 16/76, forged 12/76, cheerled 31/76, snapped 45/76.

The judge is a parameter. Table 2 reports the conduct judge (`gemini-2.5-flash`) against the human’s 304 blind labels. The two right-answer markers are licensed (κ 0.77 and 0.95), so the generation finding stands on human-validated labels. The two judgment-call markers are moderate (κ 0.68 and 0.63) and the error is one-directional: the judge over-calls departure in 20 of 25 splits, reading warmth beside an intact warning as cheerleading and missing the oblique nod to a loss; both are definitional lines, now example spans in the codebook. Two further judges from other vendors reach κ 0.77/0.78/0.58/0.45 and 0.74/0.55/0.66/0.59 on the four markers, and on every marker the best judge-human agreement is at or above the best judge-judge agreement: machine agreement is not validation. A verified quote is not a verified label either; only coding by hand puts a number on the residue. The cheap way to that number is the loop of §1: the emergent codebook, written from forty arcs in one sitting and handed closed to any of six models from three vendors, agreed with the human at κ 0.84 to 0.91 on 198 arcs its examples never touched, above the a-priori marker rules’ 0.80 on the same arcs.

4 What it is for

The same assays, unchanged, run on a deployer’s own agent: the scenes are the pressure a customer-facing agent meets, and the per-code agreement says what the judge can be trusted with. Read one column at a time, they profile a single model, the scorecard beside the coded

manner. Re-run per release with receipts, the record is a standing behavioral census a lab cannot publish about itself.¹

References

- [1] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, et al. Chatbot arena: An open platform for evaluating LLMs by human preference, 2024.
- [2] Epoch AI. Capabilities & benchmarking (epoch capabilities index). <https://epoch.ai/benchmarks>, 2026. Retrieved 2026-09-13; CC-BY.
- [3] Tapan Parikh. The one-word census: Answer-choice conformity across 44 language models, 2026.
- [4] Tapan Parikh. Conduct: a behavioral atlas of language models under pressure (study and data). <https://github.com/tap2k/modelun/tree/main/studies/conduct>, 2026.
- [5] Tapan Parikh. Every language has its own serendipity: The one-word census across languages. <https://convovo.ai/blog/every-language-serendipity>, 2026. Published 2026-07-24.
- [6] Tapan Parikh. Structured output collapses answer diversity across 44 language models, 2026.
- [7] Tapan Parikh. Tag questions and the generational reversal of sycophancy across 45 language models, 2026.

¹Data: stimuli, transcripts, scores, the label mapping, and the scripts that produce every number here are public in the `studies/cross-instrument` directory of <https://github.com/tap2k/modelun>.